

#### ④ Integration.

- Specific integral substitutions:

1)  $\frac{1}{\sqrt{a^2 - x^2}} : \cos^2 \theta + \sin^2 \theta = 1, \therefore x = a \sin \theta$

2)  $\frac{1}{\sqrt{x^2 - a^2}} : x = \cosh \theta, dx = \sinh \theta d\theta$

3)  $\frac{1}{\sqrt{x^2 + a^2}} : x = \sinh \theta, dx = \cosh \theta d\theta$

- Reduction formulae: split it into a form through which integration by parts is possible.

$$\int \sec^n x dx = \int \underbrace{\sec^{n-2} x}_{\text{diff.}} \underbrace{\sec^2 x}_{\text{int.}} dx$$

For polynomials: make sure to try and turn the differentiated term into the integrated.

- Areas of rectangles: use Riemann approximation/sum.

$$\lim_{n \rightarrow \infty} U_n - L_n = 0 : \text{simplify } U_n - L_n \text{ to } \frac{a}{n}$$

$$\therefore \lim_{n \rightarrow \infty} \frac{1}{n} = 0$$

- Pay attention to limits & width of rectangles.



## Integrals for length & surface area:

1) Arc length: Cartesian: 
$$\int_{\alpha}^{\beta} \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

Parametric: 
$$\int_{\alpha}^{\beta} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

Polar: 
$$\int_{\alpha}^{\beta} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta$$

## 2) Surface areas:

a. Cartesian: x-axis: 
$$2\pi \int_{\alpha}^{\beta} y \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

y-axis: 
$$2\pi \int_{\alpha}^{\beta} x \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

b. Parametric: x-axis: 
$$2\pi \int_{\alpha}^{\beta} y \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

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